# MediGraph: Coupling SHAP Attribution with PubMed-Grounded Retrieval for Verifiable Explanations in Drug-Target Interaction Prediction

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***Abstract*—Drug-target interaction (DTI) prediction models usually return a single probability with no mechanistic justification, and large language models can explain interactions fluently but make unsupported claims. This paper presents MediGraph, a layered explanation framework that combines SHAP attribution over molecular fingerprint and protein sequence features with a retrieval-augmented generation (RAG) module that grounds explanations in PubMed abstracts. A rule-based module links predicted interactions to target-disease associations to propose repurposing hypotheses. We evaluate the predictor on DAVIS and KIBA under random and cold-start splits, test SHAP with feature-deletion experiments, and compare ungrounded, strict-retrieval and staged-relaxation explanations on 60 curated drug-target-disease triples using an automatic claim-level judge cross-checked by a second automated judge. The predictor is competitive with classical and neural baselines under random splits and degrades when drugs or targets are unseen. SHAP-ranked deletion hurts performance more than random deletion on a random split, but we find no measurable advantage under cold-drug evaluation or over impurity-based ranking. Retrieval greatly increases the share of claims attributable to the literature, and staged relaxation with an abstain instruction declines all negative controls; because the grounded prompt restricts answers to retrieved text, we treat attributability as a design property and not as evidence of truth. The repurposing module is assessed only as an exploratory screen and does not significantly outperform random ranking. Limitations include small datasets, a small explanation test set, abstract-only retrieval, and automated judging. Code and evaluation data are released.**

***Keywords*—cheminformatics, drug repurposing, drug-target interaction, explainable artificial intelligence, PubMed, retrieval-augmented generation, SHAP**

## I. Introduction

Finding which small molecules bind which proteins is one of the slowest and most expensive steps in drug discovery. Experimental screening cannot cover the space of possible drug-target pairs, so computational drug-target interaction (DTI) prediction is used to decide where laboratory effort should go [1]. Models have moved from fingerprint-based classifiers to sequence and graph neural networks [2]–[4], and benchmark scores on kinase datasets are now high.

A high score does not tell a biologist why a pair was ranked highly, and a ranking nobody can interrogate is hard to defend when someone must decide whether to spend a month at the bench. Explainable AI has been proposed as a remedy for this in drug discovery [5]. Shapley-value attributions such as SHAP [6] are a common choice because they assign each input feature a signed contribution to a prediction. For molecular fingerprints, however, the output is a list of hashed bits, and even when a bit is traced back to a substructure it describes the behavior of the model rather than the biology. It cannot say whether the predicted relationship has ever been reported.

Large language models (LLMs) are the obvious tool for that second question, since they can write a readable account of why a drug might act on a target [7]. They also invent facts and references with confidence [8]. Retrieval-augmented generation (RAG) gives the model retrieved documents to work from [9], but retrieval can fail silently, for example by returning nothing for a sparsely studied pair, and a model told to use the evidence can still ignore it. Whether grounding actually helps in a given domain has to be measured [10].

This paper studies that question for DTI prediction. MediGraph pairs a fingerprint-based predictor with two explanation layers: SHAP attribution for structural drivers, and a PubMed-grounded generator that must tie each claim to a retrieved abstract or state that no evidence was found. Predicted interactions are further connected to target-disease associations to propose repurposing hypotheses [11]. We do not claim state-of-the-art prediction accuracy. The aim is a pipeline whose predictions, structural attributions, and literature claims can each be checked.

The main contributions are:

1) A two-layer explanation design that couples SHAP attribution on drug and protein features with retrieval-grounded literature explanations.
2) An evaluation protocol for explanation quality covering retrieval failure, claim-level support against retrieved evidence, citation validity, and abstention on cases with no evidence, using an automatic judge cross-checked by a second automated judge.
3) A comparison of ungrounded, strict-query, and staged-relaxation retrieval variants on a curated set of drug-target-disease triples.
4) A benchmark study of the predictor on DAVIS and KIBA under random and cold-start splits, with perturbation-based tests of SHAP faithfulness and stability.
5) A rule-based repurposing module with traceable evidence chains, assessed only as an exploratory retrospective screen, and an open implementation.

Section II reviews related work. Section III describes the system. Section IV details the experimental setup, Section V

reports results, Section VI discusses limitations, and Section VII concludes.

## II. Related Work

### A. Drug-Target Interaction Prediction

Early DTI methods used molecular fingerprints and protein descriptors with kernel methods or tree ensembles, and these remain strong when data are limited [1]. DeepDTA learns from raw SMILES strings and protein sequences with convolutional networks [2]. GraphDTA represents the drug as a molecular graph [3], MolTrans applies transformer encoders to substructure tokens [12], and DrugBAN adds a bilinear attention map between drug and protein substructures, with domain adaptation for out-of-distribution pairs [4]. The DeepPurpose library packages several of these encoders behind one interface [13]. Most results are reported on DAVIS [14] and KIBA [15]. Random splits let the same drug or protein appear in training and test data and are known to give optimistic estimates, so evaluation on unseen drugs or targets is more realistic [16]. We therefore report cold-start splits alongside random ones. We do not re-run these published deep models here, and our comparison is limited to the baselines described in Section IV-C.

### B. Explainability in Drug Discovery

Explainable AI methods for drug discovery include feature attribution, attention inspection, and substructure-level analysis [5]. SHAP [6] and its exact algorithm for tree ensembles [17] are widely used because they apply to any tabular representation. Attention weights in models such as DrugBAN offer interpretation inside the architecture [4]. These methods describe what drives a model's output. None of them check whether the resulting relationship is supported by the literature, which is the gap our second layer addresses.

### C. Retrieval-Augmented Generation in Biomedicine

RAG conditions a generator on retrieved passages [9], and recent surveys describe the design space [18]. Domain-specific language models such as BioGPT improve biomedical generation [19], yet hallucination remains a documented risk [8]. Benchmarking in medical question answering shows that retrieval helps in some settings and that results depend on the corpus, retriever, and model [10]. Scoring explanations at scale often relies on an LLM as judge, whose agreement with humans is good but imperfect and which can carry systematic bias [20]. We therefore cross-check the judge with a second automated judge and state its limits; independent human annotation is left for future work.

### D. Computational Drug Repurposing

Repurposing seeks new indications for approved drugs and can shorten development time [11], [21]. Computational approaches are commonly checked retrospectively against curated sets of known approved and failed drug-indication pairs, such as repoDB [22]. In this paper the rule-based module is presented as a hypothesis generator and illustrated with examples. It is not validated as a predictor of repurposing success and is not a clinical decision tool.

## III. System and Methodology

### A. Overview

MediGraph takes a SMILES string and a protein sequence and returns a binding probability, a structural explanation, and a literature-grounded rationale (Fig. 1). Feature extraction, prediction, and SHAP run offline in a Python pipeline. Retrieval and generation are served by a REST backend. Repurposing candidates and predictions are stored in a cloud database that feeds an interactive graph interface.

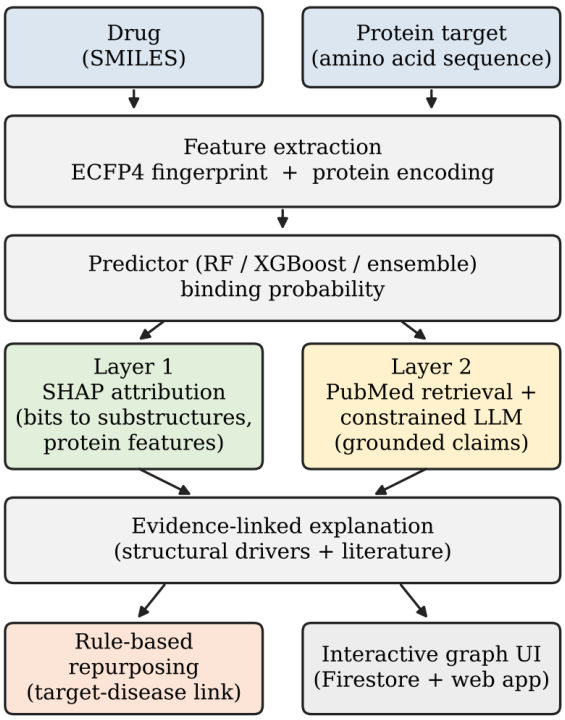



Fig. 1. MediGraph architecture. Predictions feed two explanation layers, whose outputs are combined and passed to the repurposing module and the interface.

### B. Molecular and Protein Representations

*Drug features.* SMILES strings are parsed and sanitized with RDKit [23]; molecules that fail are removed. Each valid molecule is encoded as an ECFP4 fingerprint [24], a circular fingerprint of radius 2 hashed into a 1024-bit vector.

*Protein features.* The baseline is amino acid composition (AAC), a 20-dimensional vector whose element for residue type i is

$$f_i = n_i / L \quad (1)$$

where $n_i$ is the count of residue i in a sequence of length L. AAC discards residue order, so we compare it with dipeptide composition (400 dimensions), composition-transition-distribution (CTD) descriptors (147 dimensions), and mean-pooled embeddings (320 dimensions) from the 8M-parameter ESM-2 protein language model [25] (checkpoint esm2_t6_8M_UR50D). Sequences longer than 1022 residues are truncated to the model limit. For the main benchmark, the protein vector is the concatenation of AAC, dipeptide, and CTD features (567 dimensions), and a drug-target pair is the concatenation of the 1024-bit drug vector and this protein vector (1591 dimensions). ESM-2 features are used only in the protein-feature ablation.

### C. Predictive Models

We train logistic regression, a linear support vector machine with Platt-scaled probabilities (via three-fold calibration), a random forest [26] with 100 trees, and gradient-boosted trees [27] with 100 trees, depth 5, and

learning rate 0.1. Class imbalance is handled with balanced class weights, or with positive-class scaling for boosting. Hyperparameters are fixed at these values and are not tuned. The validation split is used only for ensemble weighting and selecting the best epoch of the neural baseline. The weighted ensemble combines the random forest, the linear SVM, and gradient boosting. With $AUPR_m$ the validation average precision of model m,

$$w_m = exp(5 \cdot AUPR_m) / \Sigma_k \, exp(5 \cdot AUPR_k) \quad (2)$$

$$P(y=1|x) = \Sigma_m \, w_m \, P_m(y=1|x) \quad (3)$$

As a neural reference we train a two-tower multilayer perceptron (MLP) on the same drug and protein feature vectors, for 12 epochs with Adam at a learning rate of 0.001, with weight decay 1e-4, and keep the epoch with the lowest validation loss. This is not the published DeepDTA, which operates on raw SMILES and sequences with convolutional layers [2], and we do not label it as such.

### D. Layer 1: Structural Attribution

The Shapley value of feature i in feature set F is

$$\varphi_i = \Sigma_{S \subseteq F \setminus \{i\}} \, [|S|!(|F|-|S|-1)!/|F|!] \, [f(S \cup \{i\}) - f(S)] \quad (4)$$

and is computed exactly for tree ensembles with TreeExplainer [17]. For each prediction the system reports the top-k features by absolute value with their sign. Highly ranked fingerprint bits are mapped back to atom environments through RDKit's bit-info output and drawn on the molecule. A fingerprint bit marks an atom environment of radius at most two, which may be a small fragment and not a recognizable scaffold. These attributions explain the model, not the binding mechanism, and we test their faithfulness directly (Section IV-D).

### E. Layer 2: Literature-Grounded Explanation

*Query construction.* Given a drug, a target (gene symbol and protein name), and optionally a disease, queries are sent to PubMed through NCBI Entrez [28]. The *strict* variant issues one Boolean query that requires all entities together. Because such conjunctions often return nothing for sparsely studied pairs, the *staged relaxation* variant tries drug AND target AND disease, then drug AND target, then (drug AND disease) OR (target AND disease), and uses the first stage that returns records. If no stage returns records, the system returns an explicit no-evidence response.

*Retrieval.* Titles, abstracts, and PubMed identifiers (PMIDs) of the top three records are fetched, and all sections of structured abstracts are parsed. Every query result is cached with a timestamp so that experiments can be replayed.

*Constrained generation.* The context and the SHAP summary go to a Google Gemini Flash-class model (gemini-3.6-flash, free tier) at temperature 0. In the grounded condition the prompt requires that every claim be drawn from the retrieved text and cite its PMID, and the model must state that evidence is missing instead of filling the gap from general knowledge. Whether these constraints improve explanation quality is tested, not assumed (Section V-D). Because the grounded prompt forces the answer to stay within the retrieved text, support against that text is partly a property of the prompt, and we report it as such.

### F. Rule-Based Repurposing

A predicted pair (D, T) with probability p is kept if $p \geq 0.45$. For each disease d linked to T in the knowledge base, the pair is skipped if d is already an approved indication of D. Otherwise the candidate score is

$$s(D,T,d) = min(1, \, p \cdot \beta) \quad (5)$$

where $\beta = 1.1$ if T is a clinically validated target and 1 otherwise. Candidates with $s \geq 0.50$ are retained. The thresholds and β are heuristic choices, not fitted values. Each candidate stores its chain drug, target, disease so that a reviewer can trace why it was proposed.

### G. Implementation

Predictions, targets, diseases, and candidates are stored in Google Cloud Firestore. Only the offline pipeline, running with administrator credentials, can write prediction collections; users have read access and manage only their own saved graph views. A FastAPI service performs retrieval and generation on request. A React application (built with Vite, in JavaScript) draws the network on an HTML5 canvas and the drug-target-disease network and shows the SHAP summary and the literature rationale in a side panel. The interface is a delivery vehicle for the method and is not itself evaluated.

## IV. Experimental Setup

### A. Datasets

Predictive performance is measured on DAVIS [14] and KIBA [15], loaded through the Therapeutics Data Commons [29]. DAVIS affinities are binarized at $Kd \leq 100$ nM ($pKd \geq 7$). KIBA pairs are labeled positive when the score is at least 12.1. We confirmed the direction of the KIBA scale by matching pairs shared with DAVIS: higher KIBA values correspond to lower Kd (Spearman correlation of about −0.6 on the matched pairs we re-checked). For tractability, KIBA is subsampled to 20,000 pairs, stratified by label. Table I summarizes the data.

TABLE I
DATASET SUMMARY

| Dataset | Drugs | Targets | Pairs | Positive rate |
|---|---|---|---|---|
| DAVIS | 68 | 379 | 25,772 | 7.2% |
| KIBA (subsample) | 2,068 | 229 | 20,000 | 21.0% |

### B. Splits and Protocol

We use random (80/10/10), cold-drug, cold-target, and cold-both splits, each repeated over five seeds. In the cold splits, drugs, targets, or both are partitioned 80/10/10 so that none of the test (or validation) entities appears in training, and the overlap is checked by assertion. DAVIS contains only 68 drugs, so a cold-drug test set holds about 8 drugs, and a cold-both test set on DAVIS holds 312 pairs with only 17 positives. Cold-start results on DAVIS therefore carry high variance. DAVIS also includes mutant and phosphorylated kinase variants, so a cold-target split can still contain closely related sequences in training. We report AUROC and AUPR as primary metrics because positives are rare (the AUPR of a random classifier equals the positive rate), as mean and standard deviation over five seeds. With five seeds, paired comparisons are exploratory, and we do not report formal significance claims. F1 and MCC, computed at a threshold

tuned on the validation split, are released with the code but are not used for comparison.

### C. Baselines

Baselines are logistic regression, a linear SVM, a random forest, gradient-boosted trees, and the two-tower MLP, all on the same features and splits. We did not run published deep DTI models (DeepDTA, GraphDTA, DrugBAN, MolTrans), so claims about competitiveness refer only to this baseline set.

### D. SHAP Evaluation

Faithfulness is tested on the random forest for DAVIS under the random and cold-drug splits. Features are ranked by mean absolute SHAP value on a training subsample (300 pairs), and the top-k features (k = 5, 10, 20, 50) are masked in every test sample, either by setting them to zero or to their training mean. The k = 0 baseline and all masked runs use the same fitted model and the same test rows. We compare against k randomly chosen features, averaged over 20 draws per seed, and against k features ranked by impurity (Gini) importance, since the latter is free and would make SHAP unnecessary if equally faithful. We report AUPR over five seeds. Stability is measured on models trained on five bootstrap resamples of the training data, using the Jaccard overlap of top-k sets and the Spearman correlation of mean absolute SHAP values over the top 100 features, over all ten resample pairs.

### E. Explanation Evaluation

*Triples.* We built 60 drug-target-disease triples in three categories: 25 well-supported relationships drawn from approved indications and benchmark positives [30], [31], 20 weakly documented pairs, and 15 negative controls for which PubMed returns no record mentioning both drug and target. A negative binding label alone does not imply that no paper discusses the pair, so controls are screened by search count.

*Conditions.* C0 is LLM-only generation without retrieval, and C0b is a variant whose prompt explicitly permits an insufficient-evidence answer. C1 is the strict-query RAG described above, with a fallback to general knowledge when nothing is retrieved. C2 is staged relaxation with an explicit abstain instruction. All conditions use temperature 0.

*Metrics.* Each explanation is decomposed into atomic claims, and each claim is labeled supported, contradicted, or unverifiable against the same union of retrieved abstracts for all conditions. This measures whether a claim can be attributed to the retrieved text. It does not establish that the claim is true, because a correct claim absent from the abstracts is labeled unverifiable. We also report answer relevance, citation validity (the share of cited PMIDs that were retrieved), the rate of fabricated PMIDs (cited identifiers that do not exist), citations made without any retrieval, empty-retrieval rate, and abstention on controls. The primary judge is gemini-3.1-flash-lite, from the same vendor as the generator. Its reproducibility is checked by a second automated judge on 120 randomly sampled claims (30 per condition), with percent agreement and Cohen's kappa [32]. Confidence intervals are 95% cluster-bootstrap intervals that resample triples, not claims.

### F. Repurposing

The repurposing module is evaluated only as an exploratory retrospective screen. For each of ten known repurposed drugs, the true target competes with 100 length-matched decoys (human non-kinase proteins for nine pairs and DAVIS kinases for one), and a random forest trained on DAVIS ranks the 101 candidates. We report mean and median rank, mean reciprocal rank (MRR), and Hits@k with bootstrap intervals over the ten drugs, against the expectation under random ranking from 1,000 permutations.

## V. Results

### A. Predictive Performance

Table II reports DAVIS under the random and cold-drug splits, and Table III compares the random forest, the ensemble and the MLP across all splits and both datasets; Fig. 2 plots all models. Under the random split on DAVIS the random forest (AUROC 0.901) and the weighted ensemble (0.896) are within 0.005 of each other, and both exceed the MLP and the linear SVM. On DAVIS the ensemble does not beat the random forest consistently: it has the higher AUROC in 2 of 5 seeds under the random split, 2 under cold-drug, 1 under cold-target and 3 under cold-both. On KIBA the ensemble is ahead of the random forest in all five seeds under the random, cold-drug and cold-target splits (by 0.019, 0.005 and 0.016 AUROC on average) and in 3 of 5 under cold-both. These margins are small, so we describe the ensemble as competitive with the random forest and not better. Performance falls sharply when drugs are unseen: DAVIS AUROC for the ensemble drops from 0.896 (random) to 0.729 (cold-drug) and 0.680 (cold-both), while the cold-target split retains 0.856. DAVIS cold-both test sets hold only 17 positives per seed, so those values are noisy. On KIBA the ensemble reaches 0.739 under cold-both. The linear models are the weakest on the DAVIS cold splits: the calibrated linear SVM drops to 0.532 AUROC under cold-target with large seed variance, whereas logistic regression retains 0.720. An earlier run of logistic regression returned constant predictions (AUROC exactly 0.5) in 12 of 40 runs, which we could not diagnose; the values reported here come from a re-run with standardized protein descriptors in which no run produced constant predictions, and the other models were unaffected. Because a DAVIS cold-drug test set contains about 8 drugs, differences of a few hundredths in these columns are within seed variation.

TABLE II
DTI PERFORMANCE ON DAVIS (MEAN ± SD, FIVE SEEDS)

| Model | Random AUROC | Random AUPR | Cold-drug AUROC | Cold-drug AUPR |
|---|---|---|---|---|
| Logistic regression | 0.835±0.008 | 0.379±0.031 | 0.671±0.046 | 0.183±0.060 |
| Linear SVM (calibrated) | 0.837±0.008 | 0.382±0.035 | 0.669±0.062 | 0.188±0.082 |
| Random forest | 0.901±0.013 | 0.531±0.035 | 0.726±0.054 | 0.243±0.083 |
| XGBoost | 0.882±0.010 | 0.471±0.022 | 0.718±0.012 | 0.215±0.073 |
| Weighted ensemble | 0.896±0.009 | 0.523±0.029 | 0.729±0.033 | 0.240±0.085 |
| Two-tower MLP | 0.888±0.011 | 0.473±0.018 | 0.698±0.062 | 0.195±0.059 |

TABLE III
AUROC ACROSS SPLITS (MEAN ± SD, FIVE SEEDS)

| Dataset, split | Random forest | Ensemble | Two-tower MLP |
|---|---|---|---|
| DAVIS Random | 0.901±0.013 | 0.896±0.009 | 0.888±0.011 |
| DAVIS Cold-drug | 0.726±0.054 | 0.729±0.033 | 0.698±0.062 |
| DAVIS Cold-target | 0.862±0.025 | 0.856±0.028 | 0.829±0.029 |
| DAVIS Cold-both | 0.667±0.052 | 0.680±0.020 | 0.626±0.059 |
| KIBA Random | 0.842±0.005 | 0.861±0.007 | 0.852±0.010 |
| KIBA Cold-drug | 0.836±0.010 | 0.842±0.009 | 0.797±0.023 |
| KIBA Cold-target | 0.767±0.014 | 0.783±0.011 | 0.782±0.028 |
| KIBA Cold-both | 0.722±0.018 | 0.739±0.039 | 0.726±0.044 |

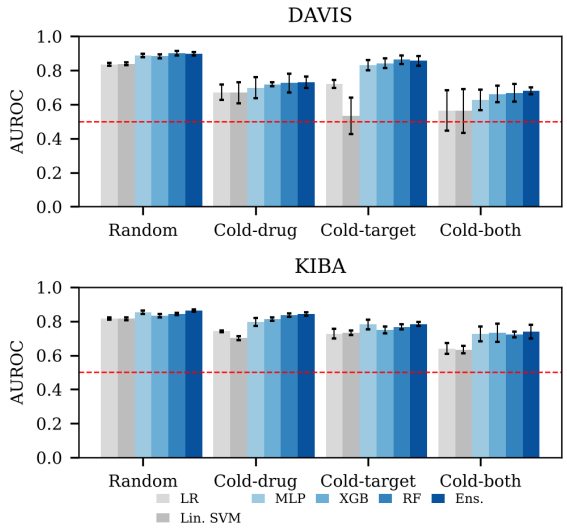



Fig. 2. AUROC (mean ± SD over five seeds) by split on DAVIS (top) and KIBA (bottom). The dashed line marks chance.

### *B. Protein Feature Ablation*

Table IV isolates the protein encoding with the random forest held fixed on DAVIS. ESM-2 embeddings give the best random-split result (AUROC 0.915±0.008) against AAC (0.871±0.008), and the AUPR gain is larger (0.564±0.027 versus 0.396±0.019). Under the cold-drug split, the same comparison is 0.734±0.061 versus 0.707±0.065, a difference smaller than the seed-to-seed spread, so we do not claim an advantage for cold-start drugs. The main benchmark uses the classical combined features, not ESM-2.

TABLE IV
PROTEIN FEATURE ABLATION ON DAVIS (RANDOM FOREST)

| Protein features | Dim. | Random AUROC | Random AUPR | Cold-drug AUROC |
|---|---|---|---|---|
| AAC | 20 | 0.871±0.008 | 0.396±0.019 | 0.707±0.065 |
| Dipeptide | 400 | 0.901±0.008 | 0.529±0.025 | 0.716±0.050 |
| CTD | 147 | 0.904±0.009 | 0.533±0.019 | 0.720±0.046 |
| ESM-2 | 320 | 0.915±0.008 | 0.564±0.027 | 0.734±0.061 |
| Combined | 567 | 0.901±0.013 | 0.531±0.035 | 0.726±0.054 |

### *C. SHAP Faithfulness and Stability*

Table V reports AUPR after masking the top-k features of the random forest on DAVIS (masking by zero) averaged over five seeds. Unmasked AUPR is 0.521±0.063 on the random split and 0.286±0.061 on the cold-drug split, and the random column averages 20 random feature sets per seed. On the random split, deleting the top SHAP features degrades AUPR far more than deleting random features (k = 50: SHAP 0.333 against 0.512 for random, lower in 5 of 5 seeds; k = 20: 0.452 against 0.517, lower in 4 of 5). Impurity-ranked deletion degrades it equally (k = 50: 0.338), so SHAP is faithful to the model on this split but is not more faithful than impurity importance. On the cold-drug split we find no reliable advantage: at k = 50 SHAP gives 0.264 against 0.280 for random, lower in only 3 of 5 seeds and within seed variation, and masking with training means instead of zeros shows no advantage either. We therefore report no measurable faithfulness advantage for SHAP under cold-drug evaluation (Fig. 3). Stability across bootstrap resamples is moderate: the mean top-20 and top-50 Jaccard overlaps are 0.497 and 0.512, and the Spearman correlation over the top 100 features is 0.439 (ten resample pairs). Hashed fingerprint bits correspond to small atom environments of radius at most two (for example a dimethylamino or an ethynyl group), so we do not interpret individual bits as pharmacophores; mapping them to chemistry is left for expert review.

TABLE V
AUPR AFTER MASKING TOP-K FEATURES (MEAN ± SD, DAVIS)

| Split | k | SHAP | Impurity | Random |
|---|---|---|---|---|
| Random | 5 | 0.519±0.108 | 0.508±0.094 | 0.520±0.062 |
| Random | 10 | 0.508±0.102 | 0.500±0.098 | 0.518±0.068 |
| Random | 20 | 0.452±0.097 | 0.426±0.086 | 0.517±0.069 |
| Random | 50 | 0.333±0.073 | 0.338±0.101 | 0.512±0.066 |
| Cold-drug | 5 | 0.289±0.058 | 0.286±0.054 | 0.286±0.061 |
| Cold-drug | 10 | 0.291±0.066 | 0.276±0.046 | 0.285±0.063 |
| Cold-drug | 20 | 0.276±0.054 | 0.283±0.077 | 0.285±0.061 |
| Cold-drug | 50 | 0.264±0.061 | 0.248±0.071 | 0.280±0.058 |

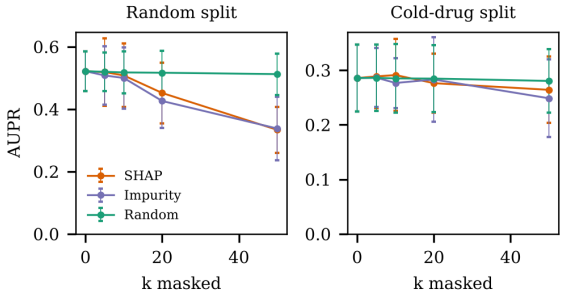



Fig. 3. AUPR on DAVIS test pairs as the top-k features are masked by zero (mean ± SD over five seeds).

### *D. Retrieval-Grounded Explanations*

Table VI compares the four conditions on the 60 triples (25 well-supported, 20 weakly documented, 15 negative controls). Intervals are 95% cluster-bootstrap intervals over triples. Strict Boolean retrieval (C1) returned nothing for 16 of 60 triples (26.7%). Staged relaxation (C2) retrieved records for all 60 (44 at the first stage, 1 at the second and 15 at the third), but the 15 third-stage retrievals were the negative controls, for which the model then abstained. Without retrieval, 98.7% of claims are unverifiable against the literature (C0), and a neutral prompt that allows abstention (C0b) raises support only to 7.5%, although it abstains on all 15 controls (C0: 2/15). C1 supports 56.9% of its claims and C2 98.6%. The C2 figure is largely a consequence of the prompt, which tells the model to answer exclusively from the retrieved text, and it is not evidence that the explanations are true. C2 abstained on 15 of 15 controls (Wilson 95% interval 79.6% to 100%) and on 1 of 45 other triples. C1 also abstained on every control, so abstention does

not separate C1 from C2 here; what separates them is coverage on the other triples, where C1 supports 171/283 claims and C2 235/239. All cited PMIDs exist (fabricated-PMID rate zero). However, in four C1 answers (triples 44, 49, 53 and 57) the model cited PMIDs although nothing had been retrieved, and a PubMed lookup returned titles unrelated to the queried drug for all six identifiers (for example, a paper on haemorrhagic diphtheria cited for mupirocin). Among answers with retrieval, every cited PMID was in the retrieved set. A second automated judge (a different Gemini model, 120 claims) agreed with the primary judge on 85.8% of claims (Cohen's kappa 0.72, substantial agreement by the Landis-Koch scale); agreement was 93.3%, 73.3%, 80.0% and 96.7% for C0, C0b, C1 and C2, and kappa is uninformative within C2 because almost all labels fall in one class. This checks the reproducibility of the judge, not its agreement with domain experts.

TABLE VI
EXPLANATION QUALITY BY CONDITION (% OF CLAIMS; EMPTY RET. = EMPTY-RETRIEVAL RATE)

| Cond. | Supported [95% CI] | Contrad. | Abstain (controls) | Empty ret. |
|---|---|---|---|---|
| C0 | 1.3 [0.0, 3.2] | 0.0 | 2/15 | n/a |
| C0b | 7.5 [3.9, 11.9] | 0.0 | 15/15 | n/a |
| C1 | 56.9 [51.5, 62.1] | 0.3 | 15/15 | 26.7% |
| C2 | 98.6 [96.9, 100.0] | 0.0 | 15/15 | 0% |

A 160-claim sample prepared for human annotation was labeled with generative-AI assistance by a co-researcher. Because this is not independent human annotation, it is not used for any result reported here, and human annotation by two or more domain experts remains future work.

### *E. Repurposing Screen*

Table VII summarizes the exploratory screen on ten drug-target pairs. The target of the true pair has a mean rank of 36.6 of 101 (95% bootstrap interval 20.8 to 53.3) against 50.9 expected for random ranking, an MRR of 0.056 against 0.051, and Hits@10 of 30% against 9.6%. No true target is ranked first or in the top five. The permutation p-value for the mean rank is 0.053, which is not significant at 0.05 with ten pairs. We therefore make no claim that the engine recovers repurposing targets. The model was trained only on DAVIS kinase inhibitors while nine of the ten targets are not kinases, so these cases lie outside its training domain. Decoys for the nine non-kinase targets are the 100 human non-kinase sequences closest in length to the true target, taken from a fixed pool file distributed with the code. Eight of the ten literature references for these cases failed automated PMID verification, so case-level citations are omitted until they are checked.

TABLE VII
EXPLORATORY REPURPOSING SCREEN (N = 101 CANDIDATES, 10 DRUGS)

| Method | Mean rank [95% CI] | MRR | Hits@10 |
|---|---|---|---|
| MediGraph | 36.6 [20.8, 53.3] | 0.056 | 30% |
| Random (1,000 perm.) | 50.9 [34.0, 67.3] | 0.051 | 9.6% |

## VI. DISCUSSION AND LIMITATIONS

A simple tree ensemble is competitive on DAVIS and KIBA but loses accuracy for unseen drugs. SHAP attributions are faithful to the model on a random split but show no measurable advantage over random deletion on cold-drug pairs, or over impurity importance. Retrieval raises the share of claims attributable to the literature, while strict queries often return nothing and an ungrounded fallback can cite irrelevant papers. Several limitations qualify these findings.

*Prediction.* The features are simple, and the benchmark data are small: DAVIS has 68 drugs and covers kinases only, so cold-drug results rest on a few drugs per split and may not transfer to other target families, and DAVIS cold-both has 17 positives per test set. KIBA was subsampled. Binarizing affinity at a threshold discards information, and kinase variants in DAVIS blur the cold-target split. Models use fixed hyperparameters without tuning, and an earlier logistic regression run failed without a diagnosed cause (Section V-A). We compare against classical models and an MLP on the same features but not against published deep DTI architectures, which may perform better under cold-start conditions.

*Attribution.* SHAP values on sparse hashed fingerprints can be unstable, and one bit can map to several substructures. A faithful attribution shows what the model uses, not what binds. Impurity importance is a free alternative that was as faithful as SHAP on the random split, so the added value of SHAP is not shown. Faithfulness was tested on DAVIS only, and agreement with chemical knowledge needs expert review.

*Retrieval and generation.* The explanation test set is small and curated by hand, so intervals are wide and category differences should be read with care. The automatic judge sees only the retrieved abstracts, so it measures attributability to that text, and a true claim that the retriever missed counts against the ungrounded condition. The grounded prompt also restricts the model to the retrieved text, so high support there is partly built in. A grounded explanation can be faithful and still not answer the question, for example when the top abstracts address a related topic. Retrieval uses abstracts only, and PubMed coverage favors well-studied pairs. Negative controls are a proxy for missing evidence, not proof of it. The judge and the generator come from the same model family, which may share biases [20]. We checked the judge only against a second automated judge, and independent human annotation is still needed. Results apply to the specific models and prompts used.

*Repurposing.* The thresholds and score factor are heuristics. The screen in Section V-E is exploratory, uses ten pairs, and shows no significant advantage over random ranking. Outputs are hypotheses for expert review and are not clinical advice.

*Future work* includes full-text retrieval, comparison with published deep DTI models, a larger retrospective repurposing evaluation with domain-appropriate decoys, annotation by independent domain experts, and prospective validation of selected candidates.

## VII. CONCLUSION

MediGraph combines SHAP-based structural attribution with PubMed-grounded explanations for DTI prediction. On DAVIS and KIBA, a fingerprint-and-descriptor tree ensemble is competitive with classical and neural baselines under random splits and loses substantial accuracy when drugs or targets are unseen. SHAP attributions are faithful to

the model on a random split but not measurably better than random or impurity-based deletion under cold-drug evaluation. Retrieval increases the attributability of generated explanations, and staged relaxation with an abstain instruction avoids empty retrieval and declined all negative controls, although the attributability figure is partly a property of the prompt. The study is limited by small benchmark datasets, a small explanation test set, the use of abstracts, an unvalidated repurposing module, and the absence of independent human annotation. We release code and evaluation data at https://github.com/ShwetaSuryawanshi-git/MediGraph so that others can repeat and extend the evaluation.

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